MILITARY SPECIFICATION

VOLTMETER, MILITARIZED, SOLID STATE, DC DIFFERENTIAL, GENERAL SPECIFICATION FOR

1. SCOPE

1.1 This specification describes a portable solid state, differential, direct current (dc) voltmeter, housed in a military combination case, and including a high voltage divider for measurements to 15,000 volts dc.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.

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MIL-P-116 - Preservation, Methods of.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-D-3464 - Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification.
MIL-N-12314 - Nozzle, Fire Hose, Water, Spray and Steam.
MIL-P-15024 - Plate, Identification -- Information and Marking for Identification of Electrical, Electronic, and Mechanical Equipment
MIL-P-19834 - Plates, Identification, Metal Foil, Adhesive Backed.
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of
MIL-M-15071 - Manuals, Equipment and Systems.
MIL-B-43291 - Box, Fiberboard, Corrugated, Double-Wall, Weather-Resistant.
MIL-I-45208 - Inspection System Requirements
MIL-C-45662 - Calibration System Requirements

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-147 - Palletized and Containerized Unit Loads, 40-Inch x 48-Inch Pallets, Skids, Runners, or Pallet Type Base.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-242 - Electronic Equipment Parts, Selected Standards.
MIL-STD-415 - Test Points and Test Facilities for Electronic Systams and Associated Equipment, Design Standard for.
MIL-STD-461 - Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-781 - Reliability Tests: Exponential Distribution.
MIL-STD-785 - Requirements for Reliability Program (for Systems and Equipment).

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PUBLICATION

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MIL-HDBK-217 - Reliability Stress and Failure Rate Data for Electronic Equipment.

DRAWING

MILITARY

RED2689314 - Power Cable Assembly for Use with Electronic Test Equipment.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article samples.- Prior to beginning production, samples shall be tested as specified in 4.5.1 (see 6.2).

3.2 General requirements.- The equipment shall meet the requirements of MIL-E-16400, except as hereinafter specified.

3.2.1 Design and construction.-

3.2.1.1 Design.- All circuitry shall exhibit sufficient stability of operating parameters to maintain calibration within the allowable tolerance without resetting any internal controls for the environmental and reliability conditions as specified herein. The circuitry of the equipment shall be designed to preclude the necessity of sorting or selecting of any part from within the stated tolerance of such part, during construction, testing, or subsequent maintenance actions.

3.2.1.2 Construction.- The instruments shall be of modular miniaturized construction wherever possible. The construction shall allow maximum accessibility to internal parts and subassemblies.

3.2.1.3 Adjustments.- Adjustments or compensating devices, capable of restoring the circuitry to the specified tolerances, shall be provided for any function or parameter in which the combined time of shelf life and usage, for a period of 6 months, causes the instrument to exceed the specification tolerances.

3.3 Functional description.- The instrument shall be an all-solid-state ±0.005 percent dc differential voltmeter operating over the range from 0 to 1100 volts dc, with a maximum null sensitivity of 0.0001 volt full scale right or left deflection. An internal precision 1100 Vdc reference supply and a Kelvin-Varley type voltage divider shall be used in conjunction with a chopper-stabilized null detector to measure unknown voltages in a true potentiometric manner over the entire 0 to 1100 Vdc range. The unit shall also be capable of operation in a direct-reading mode with ±3 percent full scale accuracy over the 0 to 1100 Vdc range. In addition to operating as a conventional differential voltmeter, the unit shall incorporate extreme isolation between inner chassis and outer combination case so that either the high or low side of the voltmeter input may be at a common mode voltage potential of ±5000 Vdc from case ground potential. The unit shall meet all specifications, regardless of the magnitude of common mode voltage up to the maximum of 5000 Vdc, and shall also withstand this potential in the non-operating mode with no resultant instrument damage or shock hazard. A separate input shall be provided to allow measurement of voltages up to 15,000 Vdc with ±0.05 percent accuracy, and with a minimum input resistance of 100 megohms.

3.4 Composition.- The instrument shall consist of the following items and parts specified herein and as required to make up a complete equipment:

(a) Precision 1100 Vdc reference supply (3.5.1)
(b) Six-digit Kelvin-Varley type voltage divider (3.5.2)
(c) High input resistance chopper-stabilized null detector (3.5.3)
(d) Regulated low-voltage power supply (3.5.4)
(e) High input resistance 15,000 Vdc voltage divider with separate input probe (3.5.5)
(f) Combination case (3.5.6)

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3.5 Performance requirements.-

3.5.1 1100 Vdc reference supply.- The 1100 Vdc reference voltage supply shall provide a precision regulated, stable voltage derived from a zener diode primary EMF. The design shall utilize only solid-state devices. Mechanical choppers shall not be used. Regulation of the 1100V reference shall be better than 2 parts per million (ppm) for a 10 percent line change from nominal. Stability shall be such that drift will not exceed 10 ppm per hour, 15 ppm per day, or 25 ppm per 60 days.

3.5.1.1 Ranges.- The 1100V reference supply shall incorporate at least four ranges of 1.1, 11, 110, and 1100 Vdc. These ranges shall provide the unit with four basic input ranges of 1, 10, 100 and 1000 Vdc, each with 10 percent overranging which shall not damage the equipment in any manner.

3.5.1.2 Polarity reversal.- Each of the ranges of the 1100V reference supply shall be reversible via a front panel switch in order to measure negative voltages without reversing input leads. Polarity switches that reverse the leads to the instrument input posts shall not be used.

3.5.2 Kelvin-Varley voltage divider.- The output of the 1100 volt reference supply, after ranging, shall be applied to a Kelvin-Varley constant-current type voltage divider. The divider shall be of such accuracy and linearity to provide an overall instrument accuracy of ±(0.005 percent of input +5 uv) from 0 to ±1100 Vdc within temperature range of 20° C to 25° C. The resistors making up the Kelvin-Varley divider shall have sufficiently low temperature coefficient of resistance to provide an accuracy of ±(0.01 percent of input +5 uv) from 0 to 1100 Vdc over the temperature range of 10° C to 40° C. Derate accuracy by 100 ppm/° C from 10° C to 0° C and from 40° C to 50° C. Absolute linearity of the Kelvin-Varley divider must be such as to provide a divider accuracy of 20 ppm of dial setting above one-tenth of full-scale.

3.5.2.1 Resolution.- Resolution of the Kelvin-Varley divider shall be sufficient to provide at least six digits of readout, which shall be equal to 1 ppm of range. A continuously variable resistor shall be used at the output of the Kelvin-Varley divider as part of the readout, to obtain a precise null.

3.5.3 Null detector.- The null detector shall incorporate high input resistance, high sensitivity, chopper-stabilization, and extreme isolation as prime parameters. The null detector shall be used in the circuit to directly compare the unknown input voltage to the Kelvin-Varley divider output voltage, and shall indicate null when these two voltages are equal. The null detector shall incorporate switching to be used in a direct, or conventional mode, indicating the magnitude of unknown within ±3 percent of range on four ranges of 1, 10, 100, and 1000V. In addition, null ranges shall be capable of use as direct-reading (±3 percent) full-scale ranges when the Kelvin-Varley divider is set to zero.

3.5.3.1 Sensitivity.- The null detector shall be designed to have sufficient sensitivity to indicate full-scale right and left deflections of at least 100 uv, 1 mv, 10 mv, and 100 mv on the 1 volt input range; 100 uv, 1 mv, 10 mv, 100 mv, and 1V on the 10V input range; 1 mv, 10 mv, 100 mv, 1V, and 10V on the 100V input range; and 10 mv, 100 mv, 1V, 10V, and 100V on the 1000V input range. The meter shall display a resolution of at least 2 uv per small scale division on the most sensitive (100 uv) null range.

3.5.3.2 Null detector input resistance.- Basic input resistance of the null detector shall be 100 megohms, so that the instrument will present 100 megohms to the unknown voltage when measured in the conventional mode on all four (1, 10, 100, 1000V) input ranges. Input resistance of the null detector shall never be less than 1 megohm ±5 percent, regardless of null range or input range.

3.5.3.3 Chopper-stabilization.- To provide suitable stability, and freedom from noise and drift, the null detector shall incorporate a suitable chopper-amplifier with synchronous demodulation. Mechanical choppers shall not be used. Chopper frequency shall be selected so that beats will not occur with line frequencies or harmonics or sub-harmonics thereof.

3.5.3.4 Isolation.- The null detector shall be provided with sufficient electrical isolation from chassis ground and other circuitry such that dc common mode rejection will be in excess of 140 db, and ac common mode rejection will be in excess of 140 db at 50, 60, and 120 Hz.

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3.5.4 Power supplies.- The instrument shall include conservatively rated low-voltage power supplies with sufficient regulation to permit unit operation as described herein to accuracies as specified herein.

3.5.5 15 kv divider.- A separate input to the unit shall be provided to measure voltages up to 15,000 Vdc with at least 100 megohms input resistance. Basic measurement accuracy at this input shall be ±0.1 percent of input, and as further described in 3.5.7. Full-scale right and left null deflections shall be 1V, 10V, 100V, 1000V, and 10,000V when using the 15,000 Vdc input.

3.5.6 Combination case.- The combination case shall be used for both environmental and electrical equipment considerations. This type enclosure shall provide protection from high humidity conditions and serve to isolate the inner chassis so that dc common mode voltages up to ±5000 Vdc may be applied to either side of the signal input. With such electrical isolation, the instrument shall have the capability to measure in a potentiometric manner, voltage differences from 0 to 1100 Vdc over a spectrum from +6100 Vdc to -6100 Vdc, with either side of the input limited to ±5000 Vdc maximum.

3.5.7 Accuracy.- Overall instrument accuracy shall at least be equal to the following:

+20° C to +25° C; ±(0.005 percent of input +5 uv) from 0 to ±1100 Vdc and ±0.05 percent of input from 1000 Vdc to 15,000 Vdc (positive or negative polarity) when using 15,000 Vdc input.

on Range I from
15° C to +35° C; ±(0.01 percent of input +5 uv) from 0 to ±1100 Vdc and ±0.1 percent of input at 15 kv input from ±1000 Vdc to ±15,000 Vdc

and on Range II from
0° C to +50° C; ±(0.015 percent of input +5 uv) from 0 to ±1100 Vdc and ±0.12 percent of input from ±1000 Vdc to 15,000 Vdc

3.5.8 Instrument input resistance.- When operating as a differential voltmeter, the input resistance shall be infinite at null over the entire 0 to 1100 Vdc input range, in both positive and negative polarities.

3.5.9 Stability of meter zero.- The combined effects of internally-generated instrument noise and line voltage variations of 10 percent shall not cause more than 2 uv meter deflection on the most sensitive (0.0001V) null range.

3.6 Electrical requirements.-

3.6.1 Warm up.- The instrument's warm up time, at temperatures down to 10° Centigrade (C), shall not exceed 30 minutes to operation within the accuracies specified herein.

3.6.2 Power.- The instrument shall be capable of operating within specifications with an input rms voltage of 115 and 230 volts, ±10 percent. The instrument shall accept and operate satisfactorily on any line voltage frequency of 50, 60, ±5 percent and 400 Hz, ±10 percent. Input power shall be approximately 25 watts.

3.6.3 Power disconnect.- A switching device shall be installed in series with the ac power input to the instrument. Both sides of the ac power input shall be fused, and both fuses shall be externally accessible. Blown fuse indicators are not required.

3.6.4 Power cable.- The power input cable shall be in accordance with type A of Drawing RED2689314.

3.6.5 Time meters.- Time meters are not required.

3.6.6 Electromagnetic interference.- The equipment shall meet all the electromagnetic interference requirements of MIL-STD-461 for class 1D equipment. The interference control plan in accordance with MIL-STD-461 shall be submitted to The Naval Electronics Command for approval within 30 days after date of contract.

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3.7 Mechanical requirements.

3.7.1 Combination case.- The unit shall be housed in a military type combination case in accordance with MIL-E-16400. The corners shall be rounded with at least 1/4 inch radius. The case shall be made of aluminum, with a gasketed cover attached by separable hinge(s) and spring-loaded pull-down type latches. A storage compartment shall be built into the case cover for accessories such as the input power cable and instruction mauual. The combination case shall be equipped with a manually-operated pressure relief valve. The combination case shall meet the submergence requirements of MIL-STD-108.

3.7.1.1 One hinged metal handle shall be provided on the front cover and on the top of the case. The handle shall have a clearance from the case sufficient to pass a block 1-7/8 by 4-3/8 inches in cross section and having the edges rounded to 15/16 inch radius. The grip portion of the handles shall have a nonmetallic material shaped to fit the hand comfortably and shall be at least equal to a 3/4-inch diameter.

3.7.2 Front panel.- The instrument front panel shall be made of a transparent material (lexan) with good optical qualities, and be able to withstand all environmental tests without deformation, cracking, or impairment of optical qualities. The panel shall be attached to the combination case with captive hardware to facilitate removal and installation of the internal unit. All panel markings shall be silk-screened on the reverse side of the front panel.

3.7.3 Isolation assembly.- To allow the measurement of signals with up to 5000 Vdc common mode voltage present, the entire inner chassis shall be supported within the combination case by an isolation assembly that allows the chassis to be at a potential of 5000 Vdc with respect to the combination case. All front panel controls shall be brought out from the inner chassis on insulated shafts. Critical attention shall be given to potential shock hazards. Any object on the front panel which could possibly represent a shock hazard including hardware, shell of the power input connector, and shields of the input probes, shall be provided with a simple means of attachment to the combination case.

3.7.4 Splashproofing of unit.- The front panel and all components mounted thereon shall be provided with suitable seals and gaskets to provide splashproof capability as defined by MIL-STD-108.

3.7.5 Dimensions.- Instruments procured under this specification shall not exceed 45 inches in girth measured around the top, bottom, and sides of the combination case. The case shall not be more than 22 inches deep with the lid in place.

3.7.6 Weight.- The total weight of the unit shall not exceed 45 pounds.

3.7.7 Cooling.- No electromechanical device such as a fan or blower shall be used in the equipment.

3.7.8 Identification of product.- Equipment shall be marked for identification in accordance with MIL-STD-130.

3.7.8.1 Identification plates.- The equipment shall carry identification plates or marking in accordance with MIL-P-15024, except as otherwise specified herein.

3.7.8.2 Overall identification.- The overall equipment identification plate shall be in accordance with MIL-P-19834 and affixed to the outside of the combination case.

3.7.8.3 Subsidiary.- The markings of subsidiary units may be engraved, cast, or etched into the material of the panel. The serial number may be engraved, cast, etched, or stamped into the material of the panel. The marking of the cover may be stamped onto the painted surface with approved ink or paint.

3.7.9 Workmanship.- Workmanship shall be in accordance with MIL-E-16400.

3.7.9.1 General.- The instruments shall be capable of meeting the design and construction requirements specified in 3.2, and shall be free from any damage or defect that could result in failure to comply with the requirements of this specification.

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3.7.9.2 Detail.- The workmanship shall be such that no evidence of any physical damage or defect shall be evident upon examination of assembly and fit, materials, parts, freedom from binding, wiring assembly (lacing) and insulation, soldering, impregnation of coils, markings and identification of parts, accessibility to interchangeable parts, freedom of parts from burrs and sharp ridges, finishes, and external alignment of component parts and assemblies.

3.7.9.3 Controls.- Operating and adjustment controls, both internal and external, shall be readily accessible, suitably arranged and of such size and construction to permit convenience and ease of operation under all service conditions. Selector switches shall have positive detent action and shall be aligned properly with their markings.

3.7.10 Signal inputs.- All inputs to the instrument (high, low, 15 kv) shall be by means of probes designed specifically to eliminate shock hazard, at the end of shielded cables which are permanently affixed to the instrument front panel. Grounding rings, attached to the cable shields, shall be provided on the probes in such a manner that they are physically between the probe tip and hand-held portion of the probe. Distance between probe tip and grounding ring shall be at least 3 inches on the 15 kv input, and 2 inches on the high and low signal inputs. The cable shall be of an EIA approved type for withstanding 15,000 Vdc. Strain reliefs shall be provided for the cable, both at the instrument front panel and at the probe. Minimum distance between probe tip and instrument front panel shall be 4 feet.

3.8 Environmental requirements.-

3.8.1 Shock, vibration, and inclination.- The equipment shall meet the shock, vibration, and inclination requirements of MIL-E-16400.

3.8.1.1 Shock.- The equipment shall withstand grade A, class I, type A shock test for lightweight equipment in accordance with MIL-S-901.

3.8.2 Temperature and humidity operating.- The equipment shall operate within the accuracies specified herein over the entire temperature range from 0° C to +50° C, with a relative humidity of 90 percent, and shall otherwise conform to class 4 of MIL-E-16400. Where reference is made to temperature and humidity ranges of this paragraph, table I shall apply.

TABLE I. Temperature and humidity ranges

Ranges	Temperature	Relative humidity
I	+15° C to +35° C	75 percent
II	0° C to +50° C	90 percent

3.8.2.1 Temperature and humidity nonoperating.- Temperatures from -62° C to +75° C shall not damage the equipment when it is not operating.

3.8.3 Altitude.- There shall be no degradation of performance when the equipment is subjected to decreased barometric pressures not less than 20.6 inches of mercury approximating an altitude of 10,000 feet.

3.9 *Parts and materials requirements.- Regardless of any other requirements, materials and parts* containing mercury shall not be used in the equipment, unless specifically required or approved by the command or agency concerned and unless the presence of mercury is noted in that requirement or approval.

3.9.1 Parts.- Unless otherwise specified herein, parts shall be selected in accordance with the highest normal standards of the industry. The requirements for use of standard parts and approval of nonstandard parts of MIL-E-16400 shall not apply. However, maximum use of parts selected from MIL-STD-242 or their commercial equivalent is recommended.

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3.10 Safety requirements.- The design of the equipment shall be planned to minimize the danger of shock or injury to operating and servicing personnel. Safety features shall be provided in accordance with the following in the precedence listed:

(a) This specification
(b) MIL-E-16400.

3.10.1 Fail safe.- The design of the equipment shall be planned to minimize undesirable secondary hazards resulting from the failure of a part or circuit.

3.11 Technical manuals.- In addition to any bulk quantity required by the contract or order, two copies of an approved technical manual prepared in accordance with MIL-M-15071 (type II production) shall be provided with each equipment.

3.11.1 Operating instructions.- One copy of an operating instruction shall be mounted in the front cover.

3.12 Maintainability.- The equipment shall be constructed to provide ease of maintenance, accessibility, and replacement of all parts.

3.12.1 Plug-in construction.- Optimum use shall be made of plug-in construction.

3.12.2 Test points.- Test points and test facilities shall be provided in accordance with MIL-STD-415.

3.12.3 Guidance meeting.- Unless otherwise directed, a guidance meeting required by MIL-STD-415 shall be held at the Naval Electronic Systems Command.

3.13 Reliability requirements.-

3.13.1 Quantitative reliability requirements.- The equipment shall have a specified (θo) mean-time-between-failure (MTBF) of 6000 hours of continuous or intermittent operation in typical laboratory usage and a θo MTBF of 1000 hours when tested as required herein.

3.13.2 Reliability program plan.- The contractor shall prepare for approval a reliability program plan in accordance with the requirements of this specification and MIL-STD-785. The reliability program plan shall describe the tasks to be performed and the procedures for conducting and controlling the reliability program. A preliminary reliability program plan shall be submitted as part of the technical proposal. A detailed reliability program plan shall be submitted to the command or agency concerned 30 days after contract award, for Government approval, with an information copy to NAVELEX Code 047.

3.13.3 Reliability prediction.- A design reliability prediction shall be performed based on stress factors and part populations technique of MIL-HDBK-217. The prediction shall be based on the operational requirements of this specification. The reliability mathematical model and reliability block diagram shall be submitted with the reliability prediction report. Mechanical parts shall be included in the prediction. The source and applicability of all failure data used for the prediction shall be documented and justified prior to use. The results of the prediction shall be compared to the specified MTBF (θo) and where the predicted values do not achieve the requirements, the contractor shall accomplish such changes in design as are necessary to increase the predicted values sufficiently to achieve the requirements. An initial report shall be submitted within 180 days after contract award to the command or agency concerned, with an information copy to NAVELEX Code 047, on the procedures, results and conclusions of the reliability prediction.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

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The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality assurance system.- The contractor shall provide and maintain an effective inspection and quality control system acceptable to the Government covering the supplies under the contract. The inspection system shall be in accordance with MIL-I-45208. A current written description of the system shall be submitted to the cognizant Government inspector, for system approval, 60 days prior to first article inspection. The written description shall include flow charts showing the complete flow of material from inspection on receipt, through all manufacturing processes, to final shipment. The various processes and inspection positions shall be clearly identified on the flow charts and cross referenced to the detailed inspection procedures and criteria of the written description of the quality control system. Any changes to the approved quality control system which might affect the degree of assurance required by this specification or other applicable documents shall be submitted to the cognizant Government inspector for approval prior to use. One copy of the flow chart shall be forwarded to the Naval Electronics Systems Command for information upon approval by the cognizant Government inspector.

4.1.2 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification may consist of (a) surveillance of the operations to determine that practice, methods, and procedures of the approved test procedures are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all the requirements of this specification (not excluding any requirement of this specification for which detailed tests are not specified herein). Failure of the contractor to correct promptly deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.2 General inspection.- Examination and testing of the equipment shall be classified as follows:

(a) Preliminary inspection
(b) First article inspection
(c) Quality conformance inspection
    (1) Production inspection
    (2) Production control inspection
(d) Environmental tests
(e) Reliability tests

4.3 Preliminary inspection.- Preliminary inspection during process of manufacture shall include such visual, electrical and mechanical examination and testing of the equipment, materials, subassemblies, parts, and accessories (including source items) as may be required to assure that the complete equipment will meet all the requirements of this specification.

4.4 Test equipment and test conditions.-

4.4.1 Test equipment.- The following items of test equipment, or units of equal or greater accuracy and range, shall be used to perform the tests specified herein. Such test equipment shall be correctly calibrated and subjected to periodic recalibration cycles as necessary to assure retention of the characteristics. When appropriate, traceability to the National Bureau of Standards shall be demonstrated. Records of calibration history shall be maintained and the latest calibration date shall be attached to each such test equipment. The contractor's calibration system shall be in accordance with MIL-C-45662.

Equipment	Model	Alternate(s)
Dc Voltage and Ratio Calibration System Consisting of	Fluke Model 7101B	
Dc Calibrator (±0.002 percent)	Fluke Model 332B	JRL DVS-106C HP 740B

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Equipment	Model	Alternate(s)
Null Detector (10 Megohm input resistance, 1 MV full scale sensitivity)	Fluke 845AB	JRL No-106 HP 419A
Kelvin-Varley Divider (0.1 ppm absolute linearity)	Fluke 720A	JRL VDR-307 ESI 900
K-V Divider lead compensator	Fluke 721A	JRL LEC-307 Gen. Res. 201
Reference Divider (10 ppm division ratio accuracy, referenced to standard cell tap)	Fluke 750A	JRL PVP-1000 and standard cell comparator ESI 1040A
High Voltage Power Supply (10 kv)	Fluke 410B	
High Voltage Divider (10 kv ±0.01 percent)	Fluke 80E-10	JRL KV-10R
Certified Standard Cell Bank		JRL SCO-106
Standard Cell (for common mode check)	Optional	Optional
Environmental Test Equipment		See MIL-E-16400

4.4.1.1 Approval.- Use of any test equipments in addition to or in place of those specified in 4.4.1 requires approval of the Government. Requests for such approval shall contain sufficient information to permit engineering evaluation without reference to any supplementary specifications or document and shall include at least the following information for each requested alternate:

(a) Functional title.
(b) Specified type(s).
(c) Requested alternate type.
(d) Paragraph numbers of tests requiring use of the specified types.
(e) Analysis of the pertinent accuracies, ranges, distortion, and so forth necessary for each test.
(f) Specifications of the requested alternate type pertinent to the test.
(g) A statement that no change in the specified test procedure is required or detailed test procedure suitable for use in this specification such as to correctly allow testing using either the specified equipment or the requested alternate.

4.4.2 Test conditions.- Unless otherwise specified herein, the instruments shall be tested under the following conditions:

(a) Temperature - Room ambient, 25° C, ±2° C.
(b) Altitude - Normal ground.
(c) Humidity - Room ambient up to 80 percent relative humidity.
(d) Power source - 115 volts rms, ±2 percent; 60 Hz, ±1 percent.
(e) Attitude - Normal operating position.

4.5 First article inspection.- First article inspection shall include all the tests as specified herein.

4.5.1 Samples.- Four samples shall be subjected to the first article tests and shall be designated models A, B, C, and D.

4.6 Quality conformance inspection.-

4.6.1 Production inspection.- Each equipment shall be subjected to the inspections of 4.7.1.

4.6.2 Production control inspection.- Production control inspection shall be conducted on samples selected in accordance with 4.6.2.1. This inspection shall consist of the tests of 4.7.2.

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4.6.2.1 Sampling.- Production control inspection shall be conducted on a sampling basis in accordance with MIL-STD-105 for general inspection level I and an AQL of 1.5 percent. The samples shall be selected from equipments which have fully passed the production inspections of 4.7.1. Each of the sample equipments shall be fully tested in accordance with the tests of 4.7.2.

4.6.2.1.1 Rejection.- In the event of failure of any equipment to pass any of the production control inspections, the entire lot shall be rejected. If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.7 Inspection procedures.-

4.7.1 Production inspection procedures.-

4.7.1.1 Preoperational inspection.- The following preoperation inspection shall be accomplished prior to other testing:

(a) Mechanical and visual.- The equipment shall be given a thorough mechanical and visual examination and test to determine that the quality of all materials and workmanship is as specified herein. Particular attention shall be given to the following:

(1) Completeness.
(2) Nameplates, identification marking, and labels.
(3) Alignment and tightness.
(4) Finish.
(5) Fit and placement of parts.
(6) Ease of operation of jacks, switches, and sliding parts.
(7) Mounting and brackets.
(8) Fastening and securing of devices or parts.
(9) Accessibility of parts.
(10) Welded and soldered joints.
(11) Forms, harnesses, and other wiring.
(12) Grounding connection.

(b) Electrical.- The equipment shall be examined and tested to confirm that the wiring is correct. Where applicable, the tests shall include the following:

(1) All intra-module wiring shall be tested to assure correctness, good electrical contact, and that undesirable shorts and grounds are not present.

(2) The module grounding system shall be tested to assure low resistance connections, proper separation of shield, signal, and framework grounds, and metal-to-metal contact for panels and components that serve as electromagnetic shields.

(3) The module shall be tested to insure against insulation breakdown or arc-over when any part of the module is subjected to an applied voltage of approximately 5 minutes duration which is either the greater of 1-1/2 times the working voltage or 500 ac volts rms.

4.7.1.1.1 Warm up.- Connect the instrument to a variable frequency power supply. Adjust the power supply to obtain 115 volts, 60 Hz. Turn instrument power switch on and allow 30 minutes for warm up.

4.7.1.1.2 Conventional mode check. Using dc calibrator, check reading of instrument in direct-reading (conventional voltmeter) mode with full scale voltages of 1, 10, 100, and 1000 Vdc. All readings must be within ±3 percent of range. Switch polarity of calibrator to check full scale left deflection ±3 percent of range. Repeat all readings taken in positive polarity with a 1 megohm resistor in series with the input. Readings must not change more than 1 percent from those recorded without series resistance.

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4.7.1.1.3 Null sensitivity.- Short the voltmeter input. On each input range and all null sensitivities, set the voltage dials to the same value as the null range setting. Meter should indicate within ±3 percent of end-scale. In the 1 Vdc range, check sensitivity in both polarities.

4.7.1.1.4 Null detector rattle.- Open the input to the unit. Check random meter needle excursions on all null ranges and input ranges. Random excursions shall not exceed 2 percent of end-scale peak-to-peak over a 3-second observation period.

4.7.1.1.5 100 uv offset.- With instrument in 10 Vdc range, and null sensitivity set to 100 uv, alternately short and open the input probes. Meter deflection shall not be more than 2 uv (2 percent of null range). With input open check deflection between positive and negative polarity. Deflection shall not exceed 2 uv.

4.7.1.1.6 Polarity reversal.- On the 1V range and using the differential mode, measure a certified standard cell with and without 100,000 ohms in series with the input, and in both polarities. All four readings shall be within 5 uv of one another on the 100 uv null range.

4.7.1.1.7 Input terminal leakage.- With the instrument input open, range switch and voltage dials set to 1,000V, null switch set to 0.01V position, observe meter offset in both polarities. Offset shall not exceed 1 mv.

4.7.1.1.8 Differential mode check.- Using the reference voltage divider (driven by the dc calibrator and standardized against the standard cell bank), measure 1,000V, 100V, 10V, and 1V in the differential mode and most sensitive null range for each input range. All readings must be within ±0.005 percent. Check the difference in null reading between a voltage dial setting of 10.000 and (9.0 plus 1V from remaining digits). Difference shall not exceed 70 uv as observed on the 100 uv null range. Measure a standard cell on the 10V range; reading shall not exceed ±0.005 percent. Measure 5 Vdc from the dc calibrator on the 10 Vdc range; error shall not exceed ±0.005 percent.

4.7.1.1.9 Off-null accuracy.- In the differential mode measure 500 Vdc from the dc calibrator. In the most sensitive (0.01V) null position, change the calibrator setting by exactly 1 mv. The instrument shall indicate between 95 percent and 100 percent full scale.

4.7.1.1.10 Off-null input resistance.- In the differential mode measure 500V from the dc calibrator with 2 megohms in series with the input. In the 0.1 V null range, change the calibrator setting by exactly 0.1V. The instrument shall indicate between 95 percent and 101 percent of full scale.

4.7.1.1.11 Kelvin-Varley linearity.- Apply 100V from dc calibrator to external Kelvin-Varley Divider Apply output of divider to input of instrument. Set instrument range to 100V and null range to 0.01V. Adjust the calibrator as necessary to effect an exact null with the external Kelvin-Varley set at unity and the instrument voltage dials set to 100.00. Turn voltage dials on the instrument one step at a time through all positions while turning the corresponding dial on the external divider simultaneously. Error as read on the meter shall not exceed ±0.002 percent of reading on voltage dials. When the first dial of the instrument is reduced to zero setting, the error shall not exceed ±200 uv.

4.7.1.1.12 Stability.- With the instrument connected to the dc calibrator and measuring 1000 Vdc in the differential mode (0.01 null sensitivity) observe stability of null over a period of 24 hours. Peak-to-peak drift shall not exceed 5 parts per million in any one hour, or 8 parts per million in 24 hours.

4.7.1.1.13 High voltage input.- Using the 10,000 Vdc power supply and 10,000 Vdc voltage divider, establish a test potential of 10,000 Vdc ±0.01 percent. Measure this potential using the 15 kv probe input of the instrument. Reading shall be within ±0.05 percent of input.

4.7.1.1.13.1 Repeat the test of 4.7.1.1.13, using 9kv, 8kv, 7kv, 6kv, 5kv, 4kv, 3kv, 2kv, 1kv. Reading shall be within ±0.05 percent for each of the above inputs.

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4.7.1.1.14 Common mode dc rejection.- Prepare a standard cell so that it may be electrically isolated to a potential of 5000 Vdc. Measure the value of the cell. Use extreme caution during the remainder of this test. Apply +5000 Vdc between the combination case and the low (common) input probe of the instrument, making certain that all potential shock hazards, including probe shields and guards, are attached to the combination case with facilities provided. Again measure the value of the standard cell. The second measurement shall not differ from the first by more than 10 uv. Repeat the entire test, using a common mode potential of -5000 Vdc.

4.7.1.1.15 Overload.- Apply 1100 Vdc to the instrument from the dc calibrator and record reading. With this potential applied constantly, switch the instrument through all combinations of range and null switch settings, including the lowest (1V) range, and most sensitive (0.0001) null range. Return the range and null settings to 1000V and 0.01, respectively. Within 30 seconds, the instrument shall read within ±0.001 percent of the original setting.

4.7.2 Production control inspection procedures.-

4.7.2.1 Weights and dimensions.- Weights and dimensions of units comprising the complete equipment shall be determined.

4.7.2.1.1 Weights.- Using weight scales, measure the weight of the equipment.

4.7.2.1.2 Dimensions.- With a length scale measure the dimensions of the equipment.

4.7.2.2 Supply line voltage and frequency.-

4.7.2.2.1 Steady state line voltage and line frequency tests.- The steady state line voltage and line frequency tests shall be as follows:

(a) Connect the power cord of the equipment to the variable frequency power supply.
(b) Set the output of the frequency power supply to 115 volts and 60 Hz.
(c) With the instrument connected to the dc calibrator, measure 1,000 Vdc in the differential mode at 0.01 null sensitivity.
(d) The results of the test in (c) above shall be reference measurements.
(e) As applicable, repeat (b) and (c) for combinations of voltage and frequency from the variable frequency power supply as specified in table II. The equipment shall be operated at least 15 minutes in each combination during which time measurements as specified in (c) above shall be taken to insure conformance.

TABLE II. Steady state voltage and frequency

Voltage (v)	Frequency (Hz)
115.0	63,57
103.5	63,57,60
126.5	63,57,60
103.5	53,47,50
115.0	53,47,50
126.5	53,47,50
103.5	440,360,400
115.0	440,360,400
126.5	440,360,400

4.7.2.2.2 Voltage transients.- Voltage transients shall be tested as follows:

(a) Connect the power cable of the equipment to the variac.
(b) Set the output voltage of the variac to 126.5 volts, 60 Hz.
(c) Perform the test specified in 4.7.2.2.1 (c).
(d) The results of the test in (c) above shall be reference measurements.

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(e) Quickly increase the output voltage of the variac to 151.8 volts, hold for approximately 2 seconds and quickly return to 126.5 volts.
(f) Repeat the test specified in (c) above.
(g) Repeat (e) and (f) four more times at 2-minute intervals.
(h) Set the output voltage of the variac to 103.5 volts.
(i) Quickly decrease the output voltage of the variac to 92.8 volts, hold for approximately 2 seconds and quickly return to 103.5 volts.
(j) Repeat the test specified in (c) above.
(k) Repeat (i) and (j) four more times at 2-minute intervals.

4.7.2.2.3 Frequency transients.- Frequency transients shall be tested as follows:

(a) Set the variable frequency power supply to 115 volts and 58.2 Hz.
(b) Decrease the power frequency quickly to 56.4 Hz, hold for approximately 2 seconds, and quickly return to 58.2 Hz.
(c) Perform the test specified in 4.7.2.2.1 (c).
(d) Repeat (b) and (c) four more times at 1-minute intervals.
(e) As applicable, repeat (a) through (d) for the frequencies as specified in table III.

TABLE III. Frequencies to be used for frequency transient tests

Starting frequency (Hz)	Transient frequency (Hz)
61.8	63.6
48.5	47.0
51.5	53.0
367.0	356.0
432.0	444.0

4.7.2.3. Total power.- Using the wattmeter, measure the input power to the equipment.

4.7.3 Environmental tests.- The following environmental tests shall be conducted on a sampling plan as specified in 4.7.3.1, except that the Government may waive this requirement in the light of evidence that the contractor's production and quality control are such as to assure manufacture of satisfactory equipment. These tests shall comprise such examinations and testing as will prove satisfactory performance during or after severe environmental conditions of immersion, shock, vibration, inclination, temperature, humidity, and salt spray. They shall also demonstrate durability and reliability.

4.7.3.1 Sampling plan.- From each lot which has passed the production inspection, a random sample of units shall be selected for environmental tests. These tests may be carried out concurrently with production control inspection on different samples from the lot being tested. The sample size and lot acceptance level shall be based upon table IV. The Government may approve the further reduction of sample sizes. When an equipment having plug-in units and accessories are procured, at least one of each type plug-in unit and accessory shall be tested in (or with) the equipment. The sample size shall be increased as necessary to meet this condition.

TABLE IV. Sampling plan for environmental tests

Lot size or monthly production	Sample size	Number of units nonconforming in any environmental test	
		Acceptance number	Rejection number
1 to 100	1	0	1
101 to 300	2	0	1
301 to 500	4	1	2
501 or more	8	2	3

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4.7.3.1.1 Rejection.- The results of each test shall be compared with the requirements of this specification. Any deviations from this specification for any such tests shall be counted as a defect and the unit shall be rejected. The contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken to correct units of product *which were manufactured with the same conditions, materials, processes, etc. If the Government inspector* does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.7.3.2 Satisfactory operation tests.- The specific test for satisfactory operation during the course of environmental tests shall consist of the repetition of the tests specified in 4.7.3.2.1 through 4.7.3.2.6.

4.7.3.2.1 Calibration check.-

(a) Set RANGE switch to 1000V
(b) Set NULL switch to TVM
(c) Set POLARITY switch to +.
(d) Set MULTIPLIER switch to X1.
(e) Set VOLTAGE DIALS to 1000.00.
(f) Apply +1000 Vdc from 332B calibrator.
(g) Test item meter shall read full scale right within ±3 percent.
(h) Set NULL switch to 0.1V.
(i) TEST item meter shall read on scale (within ±0.1V of null).
(j) Set VOLTAGE DIALS to 500.00.
(k) Apply +500 Vdc from 332B calibrator.
(l) Test item meter shall read within half scale (within ±0.05V of null).
(m) Set VOLTAGE DIALS to 000.00.
(n) Apply +0.1V from 332B calibrator.
(o) Test item meter shall read full scale right within ±3 percent.

4.7.3.2.2 100 uv rattle check.-

(a) Set RANGE switch to 10V.
(b) Set NULL switch to 100 uv.
(c) Set POLARITY switch to +.
(d) Set MULTIPLIER switch to X1.
(e) Set VOLTAGE DIALS to 000.00.
(f) Observe meter; random excursions over a 3-second period shall not exceed 1 small division.

4.7.3.2.3 100 uv offset check with controls set as above.-

(a) Short input, then remove short. Meter indication shall not change more than 2 uv.
(b) With input open, switch polarity switch to "-" position. The meter indication shall not change by more than 2 uv.

4.7.3.2.4 Null sensitivity accuracy check.-

(a) Set RANGE switch to 1V.
(b) Set NULL switch to 0.1V.
(c) Set POLARITY switch to +.
(d) Set VOLTAGE DIALS to .10000.
(e) SHORT INPUT.
(f) Meter must read full scale left ±3 percent.
(g) Switch RANGE switch thru all four (1, 10, 100, 1000V positions. Meter shall remain within 3 percent of full scale left.

4.7.3.2.5 Reference power supply ranges.-

(a) Set RANGE switch to 1V.
(b) Set NULL switch to 100 uv.

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(c) Set POLARITY switch to +.
(d) Set VOLTAGE DIALS to 1.00000.
(e) Apply +1V from 7101B Calibration System.
(f) The meter indication shall be on scale (within ±100 uv of null).
(g) Set RANGE switch to 10V.
(h) Set NULL switch to 0.001V.
(i) Apply 10V from 7101B Calibration System.
(j) Meter indication shall be on scale (within ±0.01V of null).
(k) Set RANGE switch to 100V.
(l) Set NULL switch to 0.01V.
(m) Apply 100V from 7101B Calibration System.
(n) Meter indication shall be on scale (within ±0.01V of null).
(o) Set RANGE switch to 1000V.
(p) Set NULL switch to 0.1V.
(q) Apply 1000V from 7101B Calibration System.
(r) Meter indication shall be on scale (within ±0.1V of null).
(s) Set VOLTAGE DIALS to 500.00V.
(t) Apply 500V from 7101B Calibration System.
(u) Meter indication shall be within half scale (within ±0.05V of null).
(v) Set range switch to 10 and set voltage dials to 3.00000.
(w) Apply 3000 volts from the 10,000 Vdc supply to the 15 kv divider probe.
(x) Indicated error on the 0.01 volt null range shall not exceed 0.0030 volts.

4.7.3.2.6 This test is the same as 4.7.3.2.5, except change tolerances on the following steps (use appropriate null range):

4.7.3.2.5 (f)	Change ±100 uv to ±250 uv.
4.7.3.2.5 (j)	Change ±0.001V to ±0.0025V.
4.7.3.2.5 (n)	Change ±0.01V to ±0.025V.
4.7.3.2.5 (r)	Change ±0.1V to ±0.25V.
4.7.3.2.5 (u)	Change ±0.05V to ±0.125V.
4.7.3.2.5 (x)	Change 0.0030 to 0.0033.

4.7.3.2.7 Satisfactory operation check.- Satisfactory operation check shall be made as follows:

(a) Set range switch to 10 and set voltage dials to 3.00000.
(b) Apply 3000 volts from the 10,000 Vdc supply through the 15 kv divider probe.

4.7.3.3 Temperature and humidity tests.- The following temperature and humidity tests shall be performed.

4.7.3.3.1 Nonoperating tests.- The equipment shall be subjected to the following temperature cycle and tests:

(a) Set up the equipment for test in a temperature-controlled chamber or room. Reduce chamber temperature in steps of 10° C each 30 minutes until -62° C is reached. Maintain -62° +3° C for at least 24 hours.
(b) Increase chamber temperature in steps of 10° C each 30 minutes until 75° C is reached. Maintain 75° +3° C for at least 4 hours.
(c) Reduce chamber temperature in steps of 10° C each 30 minutes until 25° C is reached. Maintain 25° +3° C for at least 4 hours.
(d) Perform the tests of 4.7.3.2.2 through 4.7.3.2.5. The accuracies of temperature and humidity (T&H) range I shall apply.

4.7.3.3.2 Operating test.- The equipment shall be capable of normal operation without alignment or adjustment of other than the operating controls throughout the temperature cycle specified hereinafter.

4.7.3.3.2.1 Conditioning.- Set up the equipment in a temperature- and humidity-controlled chamber or room. Energize the equipment. Maintain a temperature of 45° ±5° C and a relative humidity (R.H.) of 25 ±5

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percent for 8 hours. Decrease the temperature to 25° C in 2 equal steps of 30 minutes each. Immediately upon completion of the conditioning test, perform the cycling tests of 4.7.3.3.2.2.

4.7.3.3.2.2 Cycling tests.- During the following tests the relative humidity need not be controlled at temperatures below 5° C:

(a) Maintain 25° ±2° C and 25 ±5 percent relative humidity for 4 hours.
(b) Perform the tests of 4.7.3.2.1 through 4.7.3.2.5. The accuracies of temperature and humidity range I of 3.8.2 shall apply.
(c) Decrease the temperature to 15° C. Maintain 15° ±2° C and 25 ±5 percent relative humidity for 4 hours.
(d) Perform the tests of satisfactory operation tests 4.7.3.2.2 through 4.7.3.2.5. The accuracies of temperature and humidity range II shall apply.
(e) Decrease the temperature to 0° C in two approximately equal steps of 20 minutes each. Maintain 0° ±2° C for 4 hours.
(f) Perform the tests of satisfactory operation test. The accuracies of temperature and humidity range II shall apply.
(g) Increase the temperature to 35° C in steps of approximately 10° C each 30 minutes. Maintain 35° ±2° C and 25 ±5 percent relative humidity for 4 hours.
(h) Perform the tests of satisfactory operation test. The accuracies of temperature and humidity range II shall apply.
(i) Increase the temperature to 50° C in two equal steps of 20 minutes each. Maintain 50° ±2° C and 25 ±5 percent relative humidity for 4 hours.
(j) Perform the tests of satisfactory operation test. The accuracies of temperature and humidity range II shall apply.
(k) Decrease the temperature to 25° C in steps of approximately 10° C each 30 minutes.
(l) Repeat (a) through (k) for relative humidities of 50, 75, and 90 percent. The accuracies shall be as required therein except as specified hereinafter.

Relative humidity	Temperature	Accuracy T&H range (See 3.5.7)
75 percent	15° C to +35° C	I
Any	0° to 50° C	II

(m) Repeat conditioning and cycling tests (a) and (b).
(n) Perform the tests of satisfactory operation test.

4.7.3.3.3 Examination.- The equipment shall be examined in detail to detect evidence of physical degradation, such as corrosion of metal parts, distortion of plastic parts, and insufficient lubrication of moving parts. When it is necessary to replace parts to obtain satisfactory performance of the equipment the failed part or parts shall be analyzed to determine the cause of unsatisfactory operation. The results of the analysis shall be reported with the results of measurements of the equipment operating parameters. The unsatisfactory parts or material shall be replaced by adequate substitutes.

4.7.3.4 Splashproof test.-

(a) Perform the tests of 4.7.3.2.1.
(b) Mount the test item on a horizontal surface in normal operating position. The case lid shall be removed and the power cord connected.
(c) A stream of water from a 1-inch nozzle (type III per MIL-N-12314) with 10-foot head at inlet to nozzle shall be directed at the horizontal mounting surface at an angle to cause the deflected stream to strike the test item at an angle of 100 degrees from the downward vertical. It shall strike all surfaces successively for a period of 5 minutes. The nozzle shall be held 10 feet from the test item.
(d) Carefully dry the exterior of the test item, then remove from combination case. Any discernible water inside shall be cause for rejection.
(e) Perform the tests of 4.7.3.2.2 through 4.7.3.2.5.

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4.7.3.5 Altitude test.- The following altitude test shall be performed:

(a) Place the equipment in a suitable altitude chamber at 25° ±2° C.
(b) Perform the tests of 4.7.3.2.2 through 4.7.3.2.5.
(c) During the altitude tests, the equipment shall be measuring the satisfactory operation check of 4.7.3.2.7.
(d) Reduce the pressure to 20.6 inches of mercury. Maintain for 15 minutes.
(e) Observe the reading on the equipment under test. There should be no change except for allowable drift.
(f) Increase the pressure to normal ambient atmosphere. Perform the tests of 4.7.3.2.2 through 4.7.3.2.5.

4.7.3.6 Vibration test.- The following vibration test shall be performed:

(a) Using suitable canvas straps, mount the equipment (installed in its combination case but with the cover separately mounted) to the vibration table with the major horizontal axis of the equipment horizontal and parallel to the direction of vibration.
(b) Energize the equipment. After a 10-minute warm-up period, perform the tests of 4.7.3.2.2. During vibration, the equipment shall be measuring the satisfactory operation check of 4.7.3.2.7.
(c) Determine the presence of resonances in the equipment under test by vibrating at frequencies from 5 Hz to 33 Hz at a table vibratory amplitude (that is, one-half the total excursion) of 0.010 ±0.002 inch. The change in frequency shall be made in discrete steps of 1 Hz and maintained for about 15 seconds. Record frequencies at which resonances occur.
(d) Repeat (c) with 5-minute steps.
(e) Perform the tests of 4.7.3.2.2 to 4.7.3.2.5.
(f) Vibrate the equipment for a total period of continuous vibration of at least 2 hours at the resonant frequencies chosen by the test engineer. If no resonance was observed, this test shall be performed at 33 Hz. The amplitudes of vibration shall be in accordance with table V.
(g) Perform the tests of satisfactory operation test.
(h) Repeat (a) through (f) with the equipment mounted with its major horizontal axis parallel to the table and perpendicular to the direction of vibration.
(i) Repeat (a) through (f) with the equipment mounted with its major horizontal axis perpendicular to the table.
(j) Acceptability will be contingent upon the ability of the equipment to withstand tests specified above to perform its principal functions during and after vibration tests. Minor damage or distortion will be permitted during test provided such damage or distortion does not in any way impair the ability of the equipment to perform its principal functions. Non-repetitive failures of such parts as vacuum tubes, condensers, and wiring, which can be easily replaced or repaired are generally considered minor failures. If minor failures occur, the tests shall be stopped, the repair made, the entire test shall be discontinued and, after repair, the entire test shall be repeated, unless otherwise directed by the Government. The manufacturer may, at his option, substitute an entirely new equipment for retest. If this option is taken, it shall be noted in the test report furnished in accordance with 4.7.3.7.1.

TABLE V. Amplitudes of vibration

Frequency range	Table amplitude
(Hz)	(inch)
5 to 15	0.030 ±0.006
16 to 25	.020 ±0.004
26 to 35	.010 ±0.002

4.7.3.7 Shock test.- The equipment shall be subjected to the shock test, grade A, class I, type A for lightweight equipment in accordance with MIL-S-901. The following conditions shall apply:

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(a) The equipment shall be mounted with suitable canvas straps. The cover shall be removed from the case and separately mounted to the shock table.
(b) The satisfactory operation test (see 4.7.3.2) shall be performed before and after each blow.
(c) During the blows, the equipment shall be measuring the satisfactory operation check of 4.7.3.2.7.
(d) All blows of each magnitude shall be performed prior to blows of greater magnitude (i.e., the three 1-foot blows, then the three 3-foot blows, then the three 5-foot blows).

4.7.3.7.1 Vibration and shock test report.- The test report to be furnished the Government by the testing laboratory shall include detailed descriptions of any damage or malfunctioning incurred and at what stage in the tests it occurred. Photographs of physical damage shall be included. Recommendations are desired as to what corrective measure, if any, should be taken. At the discretion of the test engineer, it shall also include other pertinent information, such as the overall dimensions of the equipment, its weight, approximate location of the center of gravity, and a sketch or photograph of the methods in mounting it on the test machine.

4.7.3.7.1.1 Special shipping instruction.- The contractor shall request special shipping instructions for the specific equipments utilized in the shock and vibration tests.

4.7.3.8 Submergence test watertight.-

(a) Perform the tests of 4.7.3.2.1.
(b) Have test item stored with lid open for a minimum of 4 hours in an ambient temperature approximately 22° C above the temperature of the water to be submerged in.
(c) Close case lid, latch both latches, close manual pressure relief valve and submerge test item so there is 3 feet of water over the highest part of test item. Leave for 2 hours.
(d) Remove test item from water, carefully dry exterior, then remove from combination case. Any discernible leakage shall be cause for rejection.
(e) Perform the tests of 4.7.3.2.1 through 4.7.3.2.5.

4.7.3.9 Enclosure test.- The following enclosure test shall be performed:

(a) Repeat satisfactory operation tests (see 4.7.3.2).
(b) Mount the equipment de-energized in its combination case with cover installed in such fashion as to permit tilting of its vertical axis in any direction to the extent of 75 degrees to the tilt table directly below a hose with a 1-inch uniform diameter nozzle installed not less than 3 feet above the top of the equipment. The nozzle shall be fitted with a domestic type sprinkling faucet.
(c) The enclosure of the equipment shall be tested by subjecting it to a stream of water under a velocity head at the nozzle of 1 foot. The nozzle shall be directed downward to the test position.
(d) During the test the unit shall be tilted in all directions and subjected to the test in accordance with the following schedule:
    (1) Stream directed onto top (with table horizontal) for 15 minutes.
    (2) Stream directed for 5 minutes onto each completely closed side and each side containing controls, panels, cable entrances, louvers, and so forth (tilt the table so that the side being sprayed is 75 degrees from the horizontal).
(e) Return the equipment to the horizontal (tilt table horizontal) and repeat satisfactory operation tests. Remove the instrument from its combination case. Any leakage accumulated in pockets in the bottom or in any other part of the enclosure will constitute failure of this test.

4.7.3.10 Electromagnetic interference.- Unless otherwise specified herein, the electromagnetic interference and electromagnetic compatibility tests shall be conducted in accordance with MIL-STD-461 for the class of equipment.

4.7.4 Reliability inspection.- The reliability of the equipment shall be verified in accordance with MIL-STD-781 except as specified in subsequent subordinate paragraphs.

4.7.4.1 Reliability test plans.-

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4.7.4.1.1 Demonstration phase.- The demonstration phase of the reliability testing shall be conducted in accordance with Test Plan I of MIL-STD-781 and 4.7.4.2 on 10 equipments to be selected from the first monthly production lot.

4.7.4.1.2 Sampling phase.- Unless otherwise required by the contract or order, the sampling phase of the reliability testing shall be conducted in accordance with Test Plan V of MIL-STD-781 and 4.7.4.2 on four equipments to be selected from subsequent production lots.

4.7.4.2 Reliability test conditions and procedures.-

4.7.4.2.1 Burn-in.- Debugging and burn-in of equipments, assemblies, or parts prior to performance of the reliability testing shall be limited to the debugging and burn-in to be performed on all production equipment.

4.7.4.2.2 Preventative maintenance.- Except for normal battery replacement in equipments powered by self-contained batteries, no preventative maintenance shall be performed during the reliability test.

4.7.4.2.3 Test schedule.- The test shall consist of three 8-hour periods per day. One period shall be manned. The two remaining periods need not be manned.

4.7.4.2.4 Temperature cycling.- During the reliability test, the temperature shall be cycled in accordance with the alternative method described by paragraph 5.2.3.2 of MIL-STD-781. There shall be one complete cycle (consisting of one low and one high temperature alternation of approximately equal length) during each 8-hour period. The rate of change of temperature shall not exceed 10° C per 30 minutes. The low temperature shall be 15° C ±2° C and the high temperature shall be 50° C ±2° C.

4.7.4.2.5 Humidity.- The humidity need not be rigorously controlled during the reliability tests but the test chamber should have a relative humidity of approximately 35 percent at 25° C.

4.7.4.2.6 Vibration.- Each equipment shall be vibrated at a non-resonant frequency of approximately 25 Hz with a peak acceleration value of 1G ±10 percent for at least 80 minutes during each day of reliability testing. The temperature and humidity need not be controlled during the actual vibration period and the transport time providing the total uncontrolled time does not exceed 2 hours per day. Continuous vibration is permissible providing the other reliability test conditions are maintained.

4.7.4.2.7 Equipment on-off cycling.- The equipment shall be turned off for 30 minutes during the cooling period of each temperature cycle, except that equipment powered by self-contained batteries need not be turned off during unmanned periods.

4.7.4.2.8 Duty cycle.- The satisfactory operation tests of 4.7.3.2 shall be performed during the latter part of one low temperature alternation and during the latter part of one high temperature alternation each week. During the remainder of the on-time (including the vibration periods), the equipment shall be performing the satisfactory operation check of 4.7.3.2.7.

4.7.4.2.9 Power input voltage cycling.- Equipment powered by other then self-contained batteries shall have the input voltage sequentially cycled during each power-on period. The power input voltage shall be maintained at the nominal voltage of 3.6.2 during the first power-on period, at the high voltage of 3.6.2 during the second power-on period, and at the low voltage of 3.6.2 during the third power-on period. This sequence shall be repeated throughout the reliability test.

4.7.4.3 Failures.- Equipment failures are defined as any departure from the required performance or operation outside of the required accuracies not correctable by normal use of the operating controls.

4.7.4.3.1 Failure actions.- In the event of failure(s), failure action shall be taken as required by MIL-STD-781.

4.7.4.3.2 Corrective action.- In the event of failures, the contractor shall make any modification(s) necessary to conform with the reliability requirements. Any modification introduced into any equipments(s)

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delivered under any contract as a result of deficiencies (i.e., failure to meet the specified requirements) discovered from the reliability tests shall be included in all equipments delivered under that contract at no additional cost to the Government. Such modifications shall be included prior to delivery in all equipments not yet delivered and retroactively to those already delivered.

4.7.4.4 Reliability reports and records.-

4.7.4.4.1 Notification.- The contractor shall notify the command or agency concerned (with information copies to NAVELEX Codes 047 and 05141) of the scheduled date for initiation of reliability testing at least 30 days prior to that scheduled date.

4.7.4.4.2 Reliability test records.- The contractor shall maintain test records in accordance with MIL-STD-781.

4.7.4.4.3 Reliability test reports.- Within 20 days after completion of the reliability test, the contractor shall submit to the command or agency concerned (with information copies to NAVELEX Codes 047 and 05141) a report on the procedures, results, and conclusions of the reliability tests in accordance with MIL-STD-781, including copies of all reliability test records.

4.7.4.5 Disposition of reliability test equipment.- Upon completion of the equipment reliability tests, the units used shall be returned to the production line for any necessary refurbishing and production line final test and examinations as specified in the specific production and production control tests.

4.7.4.5.1 Special shipping instruction.- The contractor shall request special shipping instructions for the specific equipments utilized in the reliability tests.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Packaging.- Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Level A.- Voltmeters shall be packaged in accordance with Method IIb of MIL-P-116. Desiccant shall conform to MIL-D-3464, type II or III. Technical literature shall be packaged separately in a sealed, waterproof, transparent bag, minimum 4 mil, and shall not be placed within the barrier used to package any other item. Loose fill polystyrene shall not be used for packaging. Adhesives shall not be used in the closure of wraps, bags or boxes. Items comprising a complete set shall be placed in a consolidation package conforming to PPP-B-636, V2s.

5.1.1.1 Other items.- Other items, such as equipment repair parts, shall be individually packaged in transparent packages in accordance with MIL-E-17555.

5.1.2 Level C.- Each complete set shall be packaged to afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing.- Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A.- Sets packaged as specified in 5.1 shall be packed in containers conforming to PPP-B-640, class 2, or PPP-B-601, overseas type, when the gross weight exceeds 200 pounds. Closure, sealing, and reinforcement shall be in accordance with the applicable box specification, except that fiberboard boxes shall be reinforced with nonmetallic strapping. Use of loose fill polystyrene and adhesives shall be as specified in 5.1.1.

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5.2.2 Level B.- Sets packaged as specified in 5.1 shall be packed in containers conforming to MIL-B-43291 or PPP-B-640, class 2. Closure, reinforcement, use of polystyrene loose fill material and adhesives shall be as specified in 5.2.1.

5.2.3 Level C.- Sets packaged as specified in 5.1 shall be packed in containers acceptable to the common carrier and which will ensure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing or method of shipment shall comply with Uniform Freight and National Motor Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.2.4 Unitized loads.- When applicable, items packed as specified in 5.2 shall be palletized or containerized in accordance with MIL-STD-147 whenever quantities to one destination equal 50 cubic feet or more. Commercial 40-inch x 48-inch pallets may be used for Level B or C packs. Palletized loads shall be strapped to the pallet. Unitized loads shall not be subjected to rough handling tests. When applicable, items for shipboard end destinations shall be unitized by hull member.

5.3 Marking.- In addition to any special marking required by the contract or order, unit packages, consolidation packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129, except that all exterior level B pack markings shall also be waterproof.

5.4 Inspection.- Inspection of level A packages shall be in accordance with MIL-P-116.

5.4.1 First article inspection.- First article inspection, including the rough handling test, is required except where level C packaging and packing is specified. First article inspection will not be required when such a package and pack has been previously inspected and accepted for the same method for a similar item and satisfactory evidence can be furnished to the Government that items have been prepared identically with the previously approved package and pack.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing required (see 5.1 and 5.2).

6.2 First article.- Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - EC

(Project 6625-N397)

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DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

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NAVY-NAVELEX (ELEX 0517)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-V-28712(EC)

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. *IS THE SPECIFICATION RESTRICTIVE?*

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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